SQL Assignment 1

Items and Products Database — Solutions & Explanations

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Platform: Microsoft SQL Server (T-SQL)

Based on the uploaded CoachX assignment PDF (6 pages).

How to use this solution

Run the SQL script in SQL Server Management Studio (SSMS) in sections. The script creates the Brands and Products databases, creates both tables, inserts the eight sample rows shown in the assignment (PDF page 1), and then answers the Basic, Intermediate, Advanced and Super-Advanced questions listed on pages 2–6.

Important data notes

• The two source tables have no genuine common key. Joining vendor_nos to Product_Id is only a syntax demonstration and should not be interpreted as a real business relationship.
• The deletion exercise is wrapped in a transaction and rolled back by default. Replace ROLLBACK with COMMIT only if permanent deletion is intended.
• The sample table includes inconsistent-looking values (for example, Sale_Price is a per-unit price but Sales/Gross_Sales are totals). The script preserves source values except where the task explicitly requests recalculation.

Database structure

Brands.dbo.ITEMS_TABLE: Item_Id, item_description, vendor_nos, vendor_name, bottle_size, bottle_price.
Products.dbo.PRODUCT_TABLE: Product_Id, Country, Product, Units_Sold, Manufacturing_Price, Sale_Price, Gross_Sales, Sales, COGS, Profit, Date, Month_Number, Month_Name, Year.

Key SQL concepts covered

CREATE DATABASE/TABLE, INSERT, DELETE, SELECT, DISTINCT, COUNT, SUM, AVG, MAX, GROUP BY, HAVING, subqueries, joins, string functions, TOP, stored procedures, triggers, ROUND, DATEDIFF, and primary keys.

Solution script

The complete executable script is provided as a separate .sql file to preserve formatting and allow execution in SSMS. It includes comments identifying each question and the assumptions used.

Before you run it

The script drops and recreates the two assignment tables if they already exist. Back up any existing tables with those names first. It also uses CREATE OR ALTER, supported by modern SQL Server versions. Run with permission to create databases, tables, stored procedures and triggers.

Submission checklist

1. Open the .sql file in SQL Server Management Studio.
2. Run the database/table creation and insert sections.
3. Run each question section and inspect result sets.
4. Be prepared to explain why each query works and why a genuine join key is needed for meaningful joins.